

Proposed Solution Report

Date	1 July 2026
Team ID	SWTID-2026-9706
Project Title	Voice Based Concept understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	The primary objective is to develop an AI powered application that evaluates the student conceptual understanding through voice recognitions and provide feedback.
Scope	The project comprehensively allow users to upload a record speech convert it into text, compares it with reference concept using Sentence-Bert, Analysis speech, characteristic generates a score and creates a downloadable PDF Report.
Problem Statement	
Description	Students often struggle to determine whether they truly understand the concept after studying it. Existing assessment methods mainly focus on written answers rather than spoken explanations
Impact	Providing automated evaluation helps students improve concept clarity, communication skills and confidence while reducing educator manual evaluation effort.
Proposed Solution	
Approach	Develops AI application integrating Whisper Speech to text, Sentence-BERT Semantic similarity, Audio feature extractions and automated report generation.
Key Features	- Voice recording upload .

	- Concept understanding score - PDF Feedback report - Speech to text conversion - Audio feature extraction
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i3/i5 or higher
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	10 GB Free Disk space
Software		
Frameworks	Streamlit	
Libraries	Additional libraries	OpenAI Whisper, Librosa, numpy, ReportLab, Sentence-Transformers, SoundFile, Matplotlib
Development Environment	IDE	Visual Studio Code, pycharm